



NOVA SCHOOL OF
BUSINESS & ECONOMICS

Hedge Funds

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Final Project:

Statistical Arbitrage – Pair Trading Portfolio

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I. Investment Strategy Overview

The portfolio strategy develops and back tests a systematic and market-neutral statistical arbitrage strategy founded on the principles of pairs trading. The core premise is that two economically related assets sharing a long-run stochastic trend will exhibit short-term price divergences that are mean-reverting and therefore exploitable. In contrast to directional strategies, this approach aims to be completely market neutral.

A common misconception is that simultaneously taking a long position in the underpriced asset and shorting the overpriced one based on the Ordinary Least Squares (OLS) coefficient β from the regression $\log P_A = \alpha + \beta \log P_B$ guarantees market neutrality. In reality, this cointegration hedge only isolates the specific price relationship between the two assets, rather than their exposure to the broader market. To achieve true market neutrality, the position sizes in our model are explicitly adjusted using the Capital Asset Pricing Model market beta of each asset relative to a broad market proxy. The market-neutral adjustment is implemented by scaling the position in the lagged leg by the ratio of the two assets' rolling CAPM betas estimated over a 252-day window against SPY, formally $Mkt_{adj} = \left| \frac{\beta_L}{\beta_G} \right|$, clipped to $[0.5, 2.0]$ to guard against extreme estimates, and applied on top of the cointegration hedge ratio. Therefore, the systematic exposure to broad market moves can be hedged away, leaving the pure spread between the specific pairs as the sole source of returns.

The initial universe of potentially tradable pairs was formed by identifying candidate combinations of assets belonging to the same industries or acting in similar segments of a product or supply chain. The selection rationale is that firms exposed to similar economic and regulatory conditions are more likely to exhibit stable valuation dynamics. This intuition serves as a prerequisite for a durable statistical relationship and reduces the risk of spurious cointegration. The back testing period spans from January 2012 to March 2026, utilising a 65/35 percent split for the in sample and out-of-sample periods to effectively reduce overfitting.

II. Methodology

a. Cointegration Screening

The core of the individual pair selection revolves around cointegration rather than correlation. While correlation measures temporary co-movements between returns, cointegration aims to identify a long-run equilibrium between the price spread of each pair, a relationship that acts as a gravitational anchor pulling the spread back whenever it deviates. However, correlation remains a fundamental metric during the final portfolio construction phase. By selecting a basket of cointegrated pairs that exhibit exceptionally low correlation with one another, the strategy effectively minimises the overall variance of the final portfolio, harnessing diversification to further improve risk adjusted returns.

To test cointegration, candidates are subjected to a rigorous series of four statistical tests. The first is the Augmented Dickey Fuller test, which evaluates whether the individual stocks are stationary or follow a random walk. Cointegration can only be significant between two series of the same integration order, where price series are non-stationary in levels but stationary in first differences. The Augmented Dickey Fuller test is run twice per series: once on log price levels, where the null hypothesis of a unit root must be rejected, followed by log returns, where the null must be rejected for the pair to continue in the selection process.

The second step is the Engle Granger procedure, which examines whether a stationary linear combination of the two price series exists. The hedge ratio β is estimated by regressing $\log P_A$ on $\log P_B$ via OLS, and an Augmented Dickey Fuller test is then applied to the residuals. Although the Engle Granger framework is theoretically asymmetric, empirical tests conducted by swapping the independent and dependent variables yielded identical statistical conclusions, thereby verifying the robustness of the relationship.

The Johansen trace test operates within a Vector Error Correction Model framework and directly estimates the number of cointegration vectors. If the trace statistic exceeds the 95% critical value, at least one stable long run relationship is confirmed. Recognising its superior statistical power, a specific override rule was implemented: a pair passing the Johansen test with a non-trivial hedge ratio can still be admitted even if it marginally fails the Engle Granger procedure. Importantly, any admitted pair is still strictly required to clear a minimum of three out of the four total statistical criteria to proceed to the back testing phase.

The final criterion evaluates the half-life of the mean reversion relationship. Through a regression on the spread, the model measures the trading days required for the spread to cover half of the distance back to the mean. This acts as an

economic viability screener, as a pair can be statistically cointegrated but untradable if its reversion is too slow, causing carrying costs, borrowing fees, and margins to erode profits. Therefore, a strict half-life limit of 120 days was applied. This metric also calibrates key time parameters, setting the maximum holding period to 1.5 times the half-life and the Z score lookback window to $\max(252, 2.5 \times \text{half-life})$ days, ensuring the window always covers a sufficient number of mean-reversion cycles. In cases where the AR(1) regression yields no evidence of mean-reversion, the half-life defaults to 60 days to prevent the pair from being discarded on purely numerical grounds.

b. Backtesting Engine

To rigorously evaluate the trading logic, all baseline statistical parameters are calibrated strictly on the in sample period. During the out-of-sample phase, the hedge ratio is dynamically updated using a walk-forward approach every 63 days, relying exclusively on trailing data. Specifically, each recalibration uses an expanding window incorporating all available data up to $t-1$, rather than a fixed rolling window, so that the hedge ratio estimate becomes progressively more stable as the OOS period advances.

Trading signals are generated using a rolling Z score of the spread, calculated via an Exponentially Weighted Moving Average to allow faster adaptation to structural drifts. To eliminate any look-ahead bias, signals are computed on the close of day t and executed at the open of day $t+1$. A market-neutral position opens when the Z score exceeds an entry threshold of 1.5 in absolute terms, anticipating a reversion to the historical mean. The primary exit signal triggers when the Z score reverts to 0.0, capturing the full convergence of the spread. Before deploying capital, two execution filters guarantee market liquidity and prevent adverse pricing. First, a liquidity filter mandates that both assets must consistently maintain a minimum twenty-day average traded volume of 500,000\$. Second, an overnight gap filter aborts any pending entry signal if either asset opens with a price jump exceeding 3% compared to its previous close, successfully mitigating uncontrolled execution slippage.

Rather than allocating a fixed capital amount, the engine dynamically determines position sizing through volatility scaling. The algorithm normalises exposure so that every pair contributes equally to a target annualised volatility of 10%. The position size scalar is computed as the average of the volatility-scaled notional of both legs, ensuring symmetric exposure across the pair. Without this risk parity mechanism, highly volatile pairs would completely dominate the portfolio profit and loss, rendering defensive pairs virtually irrelevant and destroying cross-sectional diversification. Statistical arbitrage is exceptionally vulnerable to structural breaks. To strictly limit maximum drawdowns, a hard stop loss is enforced if the Z score reaches 4.0. An economic time stop forcibly liquidates any trade remaining open beyond 1.5 times the estimated half-life, preventing extended holding periods from accumulating excessive margin costs. Finally, a per-pair circuit breaker permanently halts new trades once the drawdown from the peak cumulative PnL of that pair exceeds 25% of allocated capital, and a broad execution filter prevents new entries when the VIX index exceeds 35. Critically, this VIX filter only blocks the opening of new positions and does not force-close existing trades, a design choice whose implications are further discussed in the Limitations section.

c. All-Star Portfolio Construction

To properly contextualise the aggregate results, it is necessary to outline the theoretical capital allocation framework. The strategy assigns a baseline notional value of 10,000 dollars per leg, resulting in a gross exposure of 20,000 dollars for each market-neutral pair. Across six admitted pairs, this translates to a theoretical deployed portfolio capital of 120,000 dollars. This baseline allocation serves as a reference scale for the volatility scaling engine, which dynamically adjusts actual exposure to maintain the 10% volatility target.

Candidate pairs are allocated to the final All-Star portfolio only if they clear a rigorous set of out-of-sample hurdles. A pair is admitted if it generates a minimum of three trades, achieves a win rate strictly above 55%, produces a positive Sharpe ratio, and maintains a minimum gross edge of 0.75x over total transaction costs. The fundamental rationale behind aggregating the surviving pairs is variance reduction through cross-sectional diversification. Because the selected pairs span different sectors and capture idiosyncratic price relationships, their trading signals trigger independently. By combining these orthogonal return streams, the aggregate portfolio achieves a substantially smoother equity curve and a significantly lower overall variance than any individual component.

d. Transaction Costs

In order to model realistic transaction costs, several components were applied. Firstly, a fixed 5bps cost is charged on every transaction. Slippage is fixed at 3bps, capturing market movements at trade execution. On top of these fees, a fixed commission of 1\$ is charged per trade. All transaction cost components are applied at both entry and exit of every trade. Finally, borrow and margin costs are plotted in the Appendix (Figure Cost Drag).

III. Results

a. Economic Rationale and Pair Performance

The final selected universe comprises six highly distinct pairs representing specific sectors: consumer staples (Kimberly Clark and Clorox), restaurants (McDonalds and Yum Brands), healthcare (Humana and Centene), oil services (Halliburton and Schlumberger), utilities (Xcel Energy and WEC Energy), and precious metals (the SLV and PSLV exchange-traded funds). By operating in identical market segments, these companies are exposed to the same macroeconomic cycles and regulatory shocks, justifying a long-run equilibrium in their prices. Critically, what underpins cointegration across all six pairs is not merely price correlation, but the presence of a shared macro factor exposure, whether driven by commodity cycles, regulatory rate-setting, or consumer spending pressure, combined with the absence of any structural reason for a permanent divergence in fundamentals. When a spread widens, there is an economic forcing mechanism that pulls it back, which is the necessary condition for a mean-reverting relationship to persist out of sample.

Analysing individual performances, the KMB and CLX pair stood out as the most performant, recording an 88% win rate over eight trades and generating an annualised return of 2.28% with a Sharpe ratio of 0.96. Conversely, the silver funds (SLV and PSLV) generated the highest trading frequency with 16 operations but produced the lowest profit margin, highlighting how highly arbitrated markets leave minimal room for net profits after execution costs. This result is consistent with the theoretical framework: the tighter and more mechanical the cointegration relationship, the more efficiently the market prices it, compressing the available edge. Overall, all six pairs demonstrated remarkable resilience out of sample, with win rates exceeding 60% and a positive gross edge.

b. All-Star Portfolio Performance

Operating within the volatility weighted structure, the aggregate portfolio generated a total net return of 7,562 dollars. The defining characteristic of this approach is its capacity to deliver consistent profits while severely suppressing variance. On an annualised basis, this translates to a return of 1.22% relative to deployed capital, paired with an exceptionally low annualised volatility of 1.07%. This combination produced an excellent Sharpe ratio of 1.14. Performance consistency over time serves as empirical proof that the risk management filters functioned as intended. As illustrated by the annual returns (*Figure 8*), the strategy remained consistently profitable across the entire out of sample window, registering returns of 0.0% in 2021, 0.1% in 2022, 1.4% in 2023, 2.2% in 2024, 1.2% in 2025, and 1.5% in 2026.

From a capital protection perspective, the maximum drawdown during the testing period was rigorously limited to just 1,611 dollars, representing a mere 1.34% of allocated capital (*Figure 8*). Absolute percentage returns appear modest exclusively because the strategy was tested without financial leverage; the exceptional quality of these risk adjusted metrics strongly suggests the portfolio is an ideal candidate for institutional leverage.

c. Risk Profile and Return Distribution

A detailed analysis of the daily profit and loss distribution confirms characteristics typical of statistical arbitrage. The standardised return distribution (*Figure 8*) exhibits fat tails, indicated by a positive excess kurtosis, signifying that extreme movements are more frequent than in a perfect normal distribution. This distribution is computed exclusively over days with active positions, excluding flat days where no trades are open, in order to accurately reflect the true shape of trading-day returns rather than diluting the distribution with structural zeros. Furthermore, the distribution is right skewed, implying the probability of achieving substantial gains exceeds that of suffering extreme losses. The strong central peak reflects the model remaining safely neutral during periods of market noise lacking clear statistical signals. Finally, the correlation matrix (*Figure 8*) validates the fundamental diversification hypothesis. Correlation values among the six pairs are almost universally close to zero, demonstrating that trading signals trigger independently. This orthogonality is the primary factor enabling the aggregated portfolio to drastically reduce overall volatility.

IV. Sensitivity Analysis

To validate the robustness of the strategy against parameter overfitting, a sensitivity analysis was conducted on the out of sample admitted pairs (*Figure 9*). For each parameter grid point, the backtest engine is re-run on the same six admitted pairs without repeating the cointegration screening, ensuring the analysis tests execution robustness rather than inadvertently re-optimising pair selection. The reported Sharpe ratio at each grid point reflects the aggregate portfolio-level metric across all six pairs. The portfolio Sharpe ratio demonstrates remarkable resilience to transaction costs, decaying linearly but remaining highly profitable even at extreme assumptions of 20 basis points. This confirms that the statistical edge is genuine and not a mere microstructure artifact. Regarding signal generation, the baseline Z score entry of 1.5 and exit of 0.0 proved optimal. Increasing the entry requirement severely degrades performance by executing too few trades, thereby destroying cross sectional diversification. Conversely, exiting prematurely before full mean reversion steadily erodes the risk adjusted returns. Finally, while extending the maximum holding period beyond 1.5 times the half-life yields marginal Sharpe improvements, it simultaneously exposes the portfolio to extended capital lockups and higher accrued margin costs, validating our baseline as the most prudent and capital efficient choice.

V. Limitations

The candidate universe was constructed from firms actively traded through 2026 with continuous histories back to 2012, excluding delistings and bankruptcies and introducing survivorship bias. While parameter selection is strictly contained to the in-sample period and the sensitivity analysis confirms robustness across key signal and cost parameters, fixed bid-ask and slippage estimates may understate true frictions in live trading, particularly during periods of elevated volatility. On that note, while the VIX filter prevents new entries during stress periods, it does not close existing positions opened prior to a volatility spike, leaving the portfolio exposed to spread divergence driven by broad market deleveraging rather than idiosyncratic dynamics. Finally, the rolling Sharpe ratio exhibits significant instability, which is structurally expected in mean-reversion strategies; negative periods typically reflect temporary spread divergence rather than model failure, as positions become more likely to converge the further the spread deviates. This instability is visualised using a 63-day rolling window annualised by $\times\sqrt{252}$ (*Figure 7*). However, this is indistinguishable in real time from a permanent structural break in the cointegration relationship, representing the core unresolvable risk of the strategy.

VI. Appendix

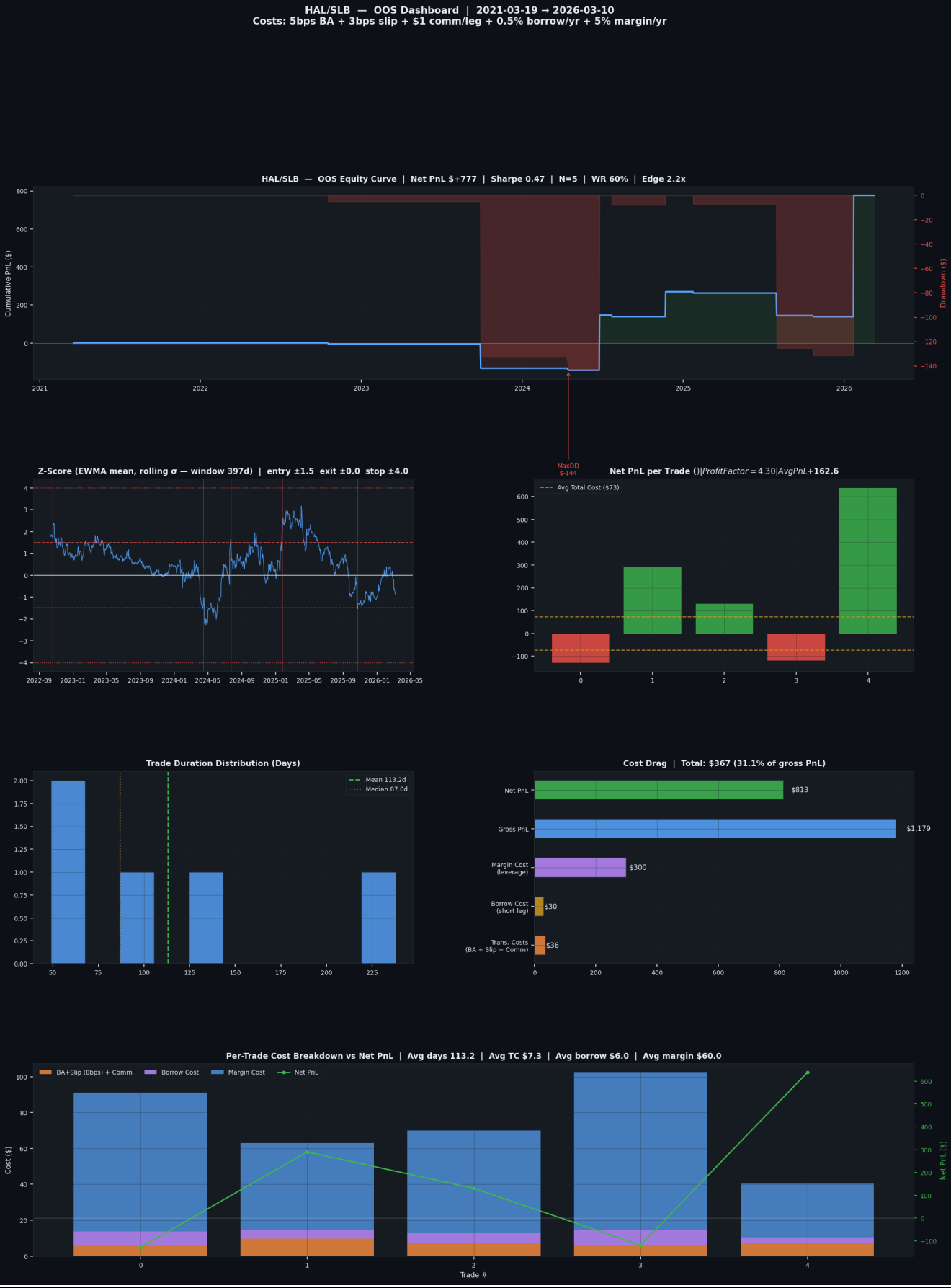


Figure 1

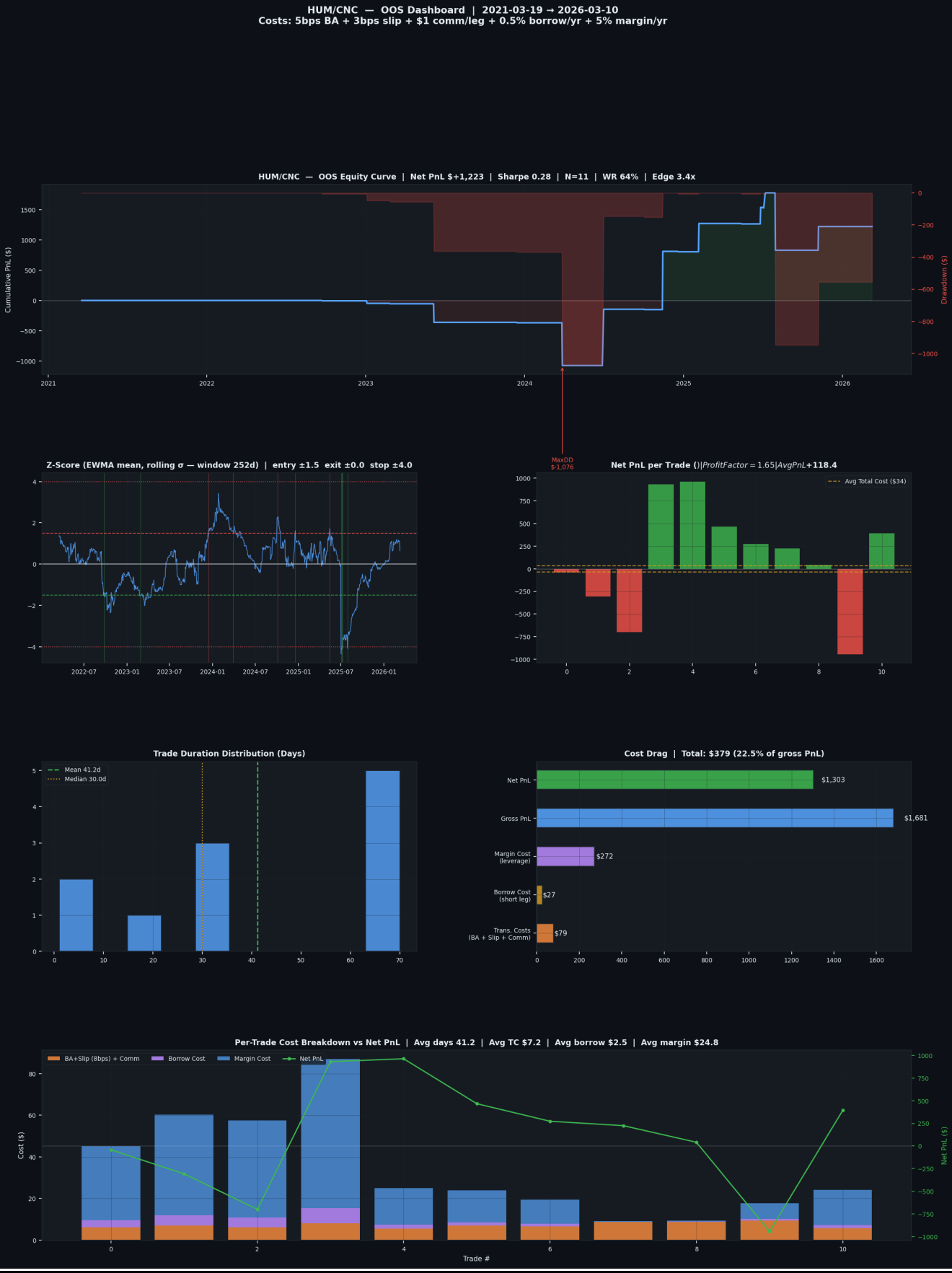


Figure 2

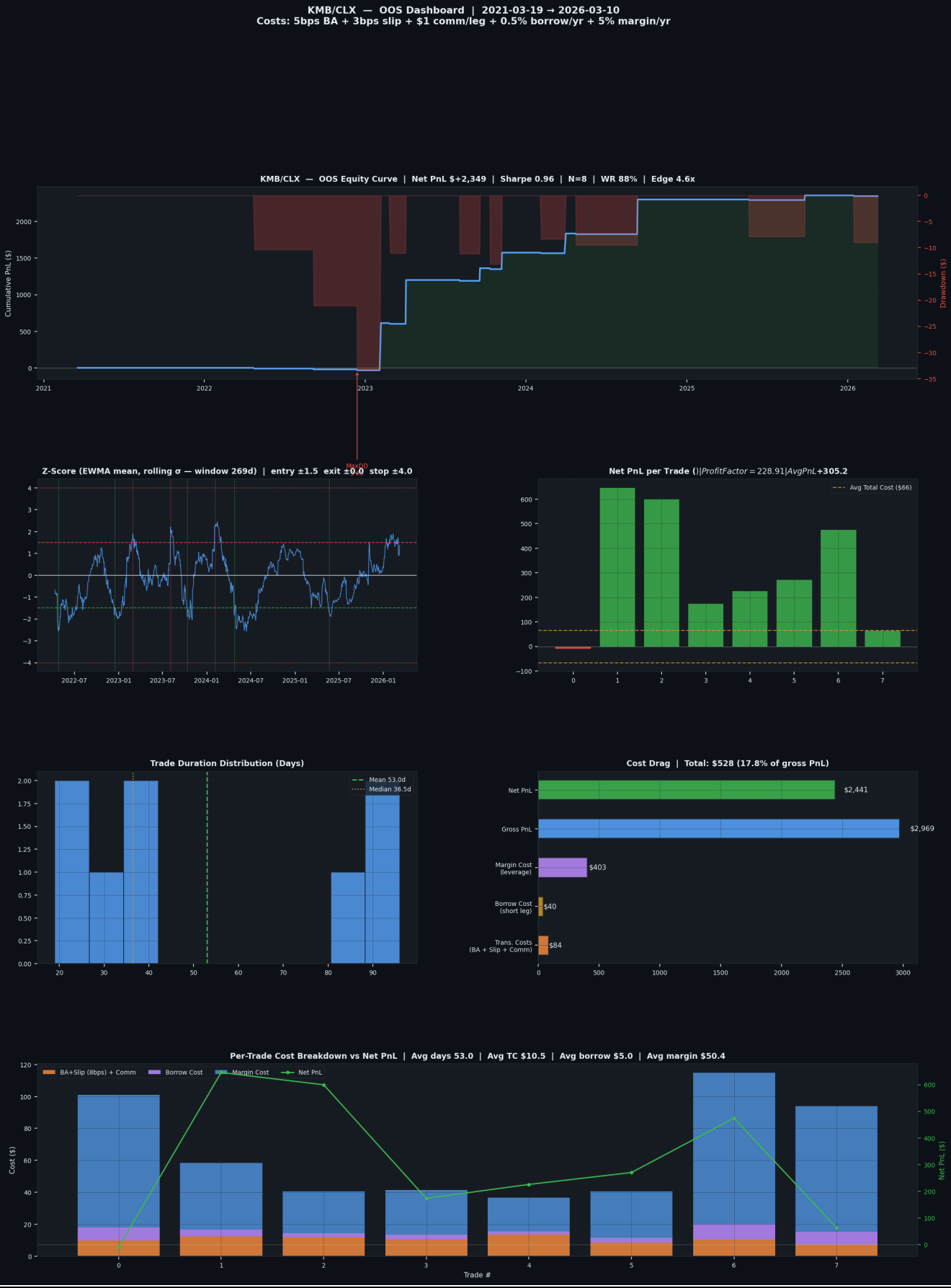


Figure 3

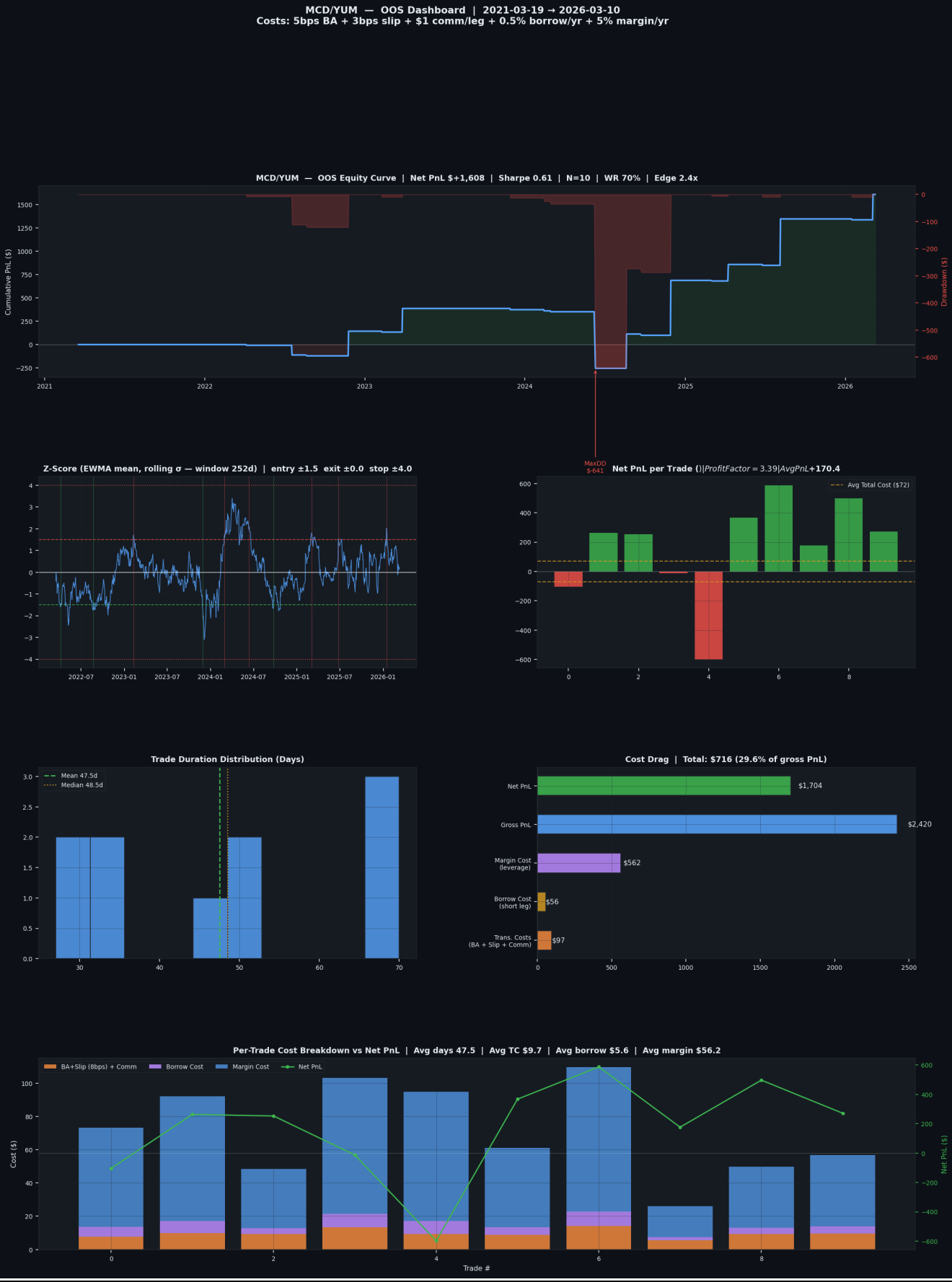


Figure 4

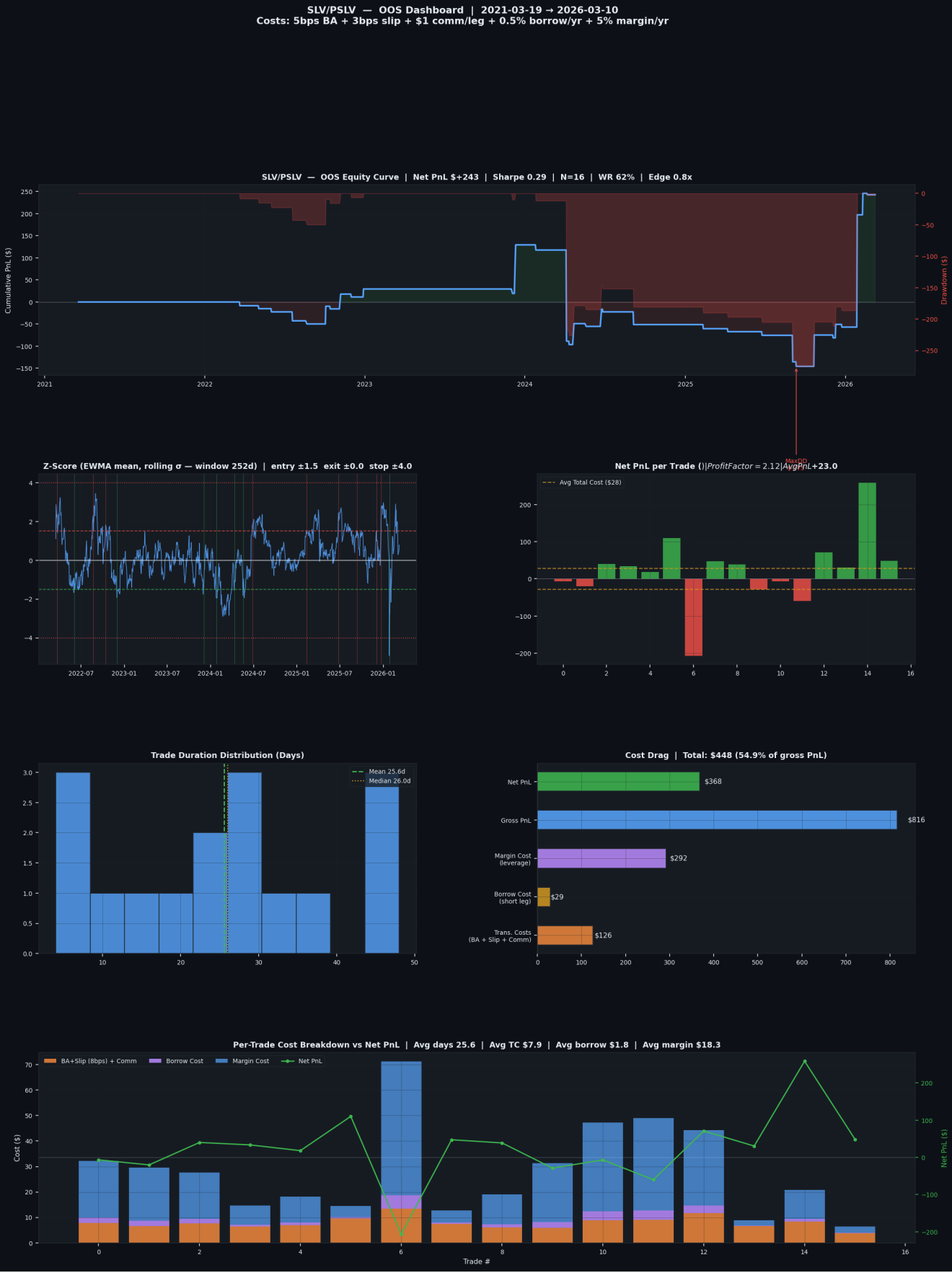


Figure 5

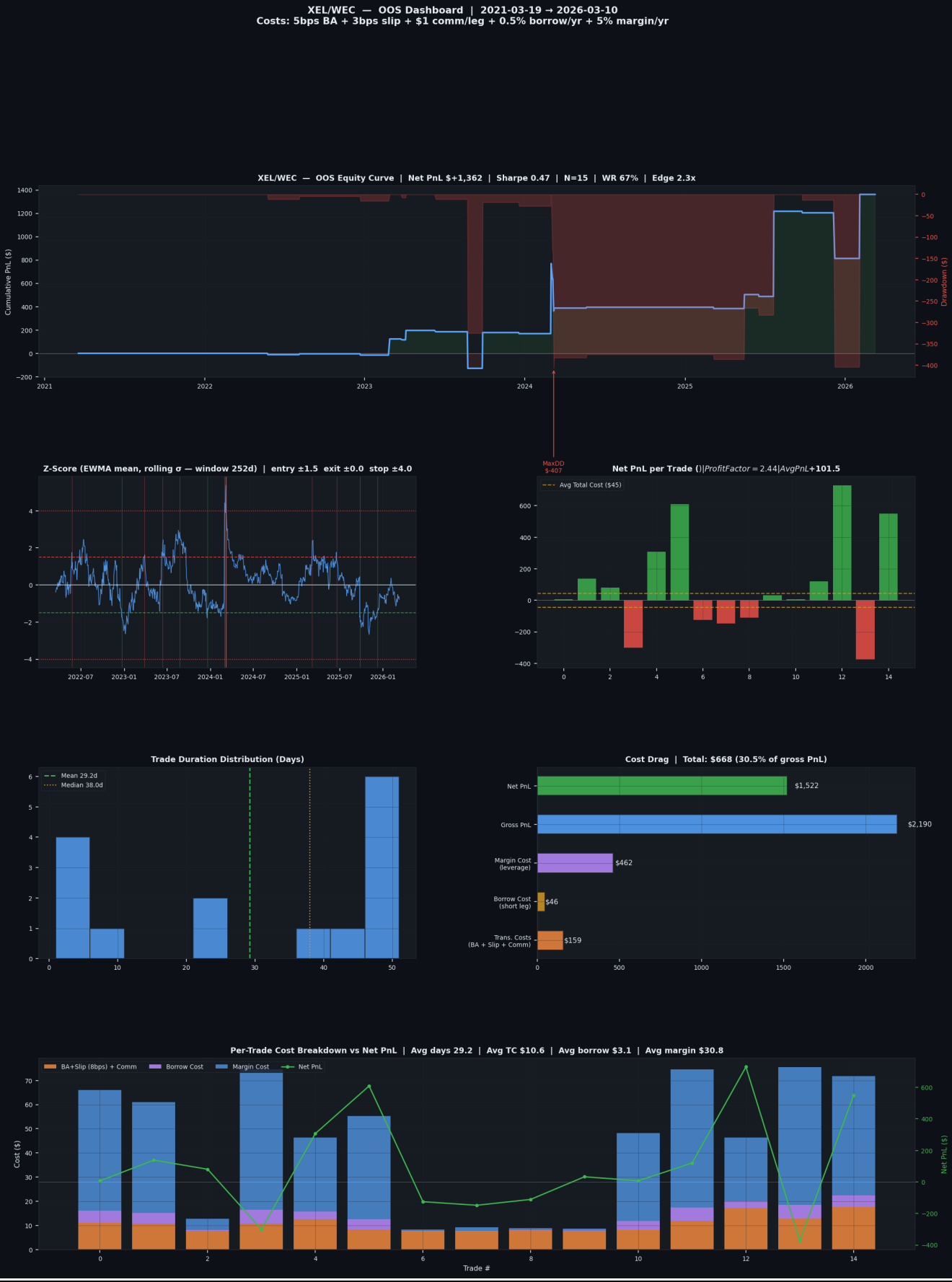


Figure 6

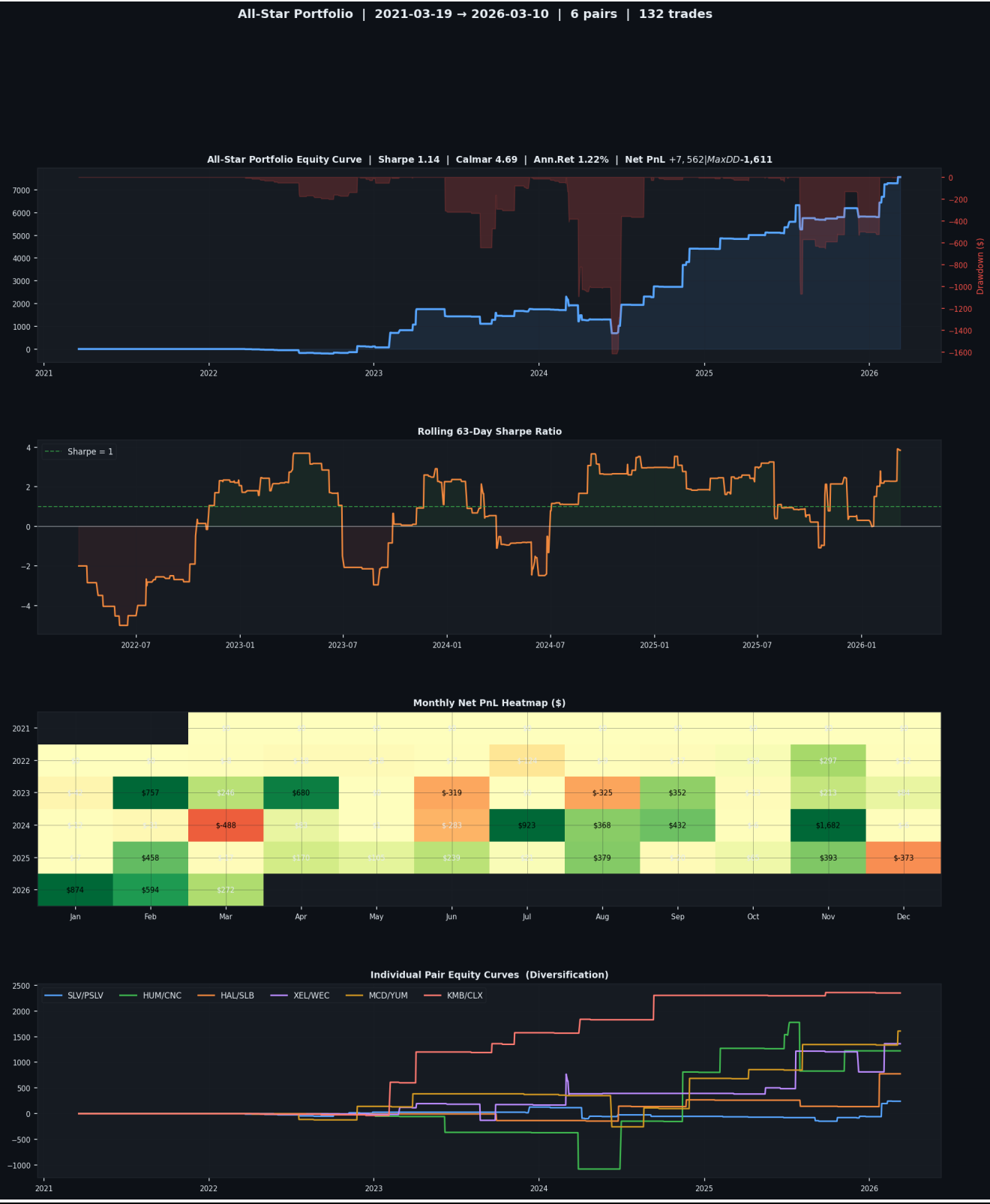


Figure 7



Figure 8



Figure 9